

What is Statistics ?

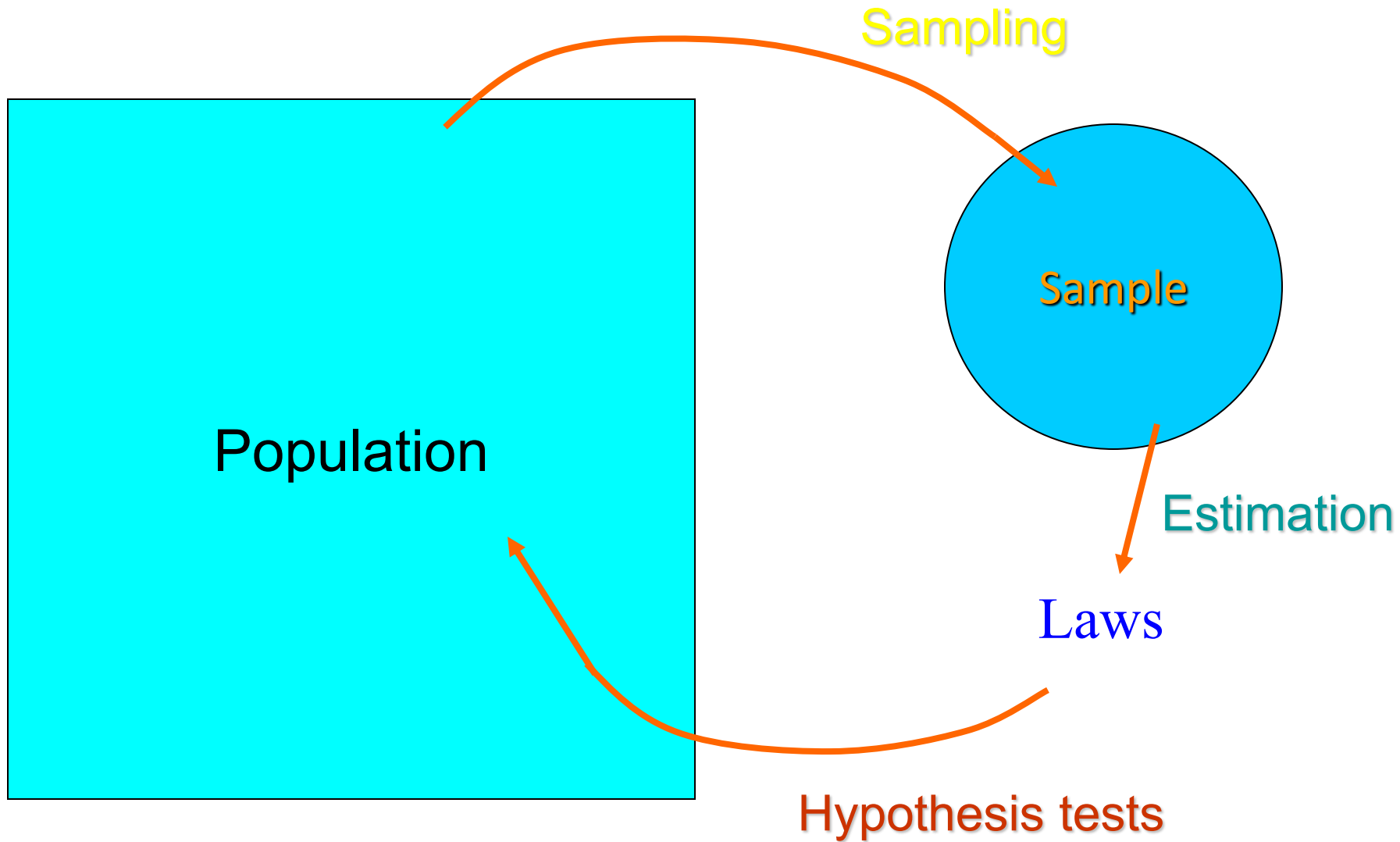
Science of investigating population's laws

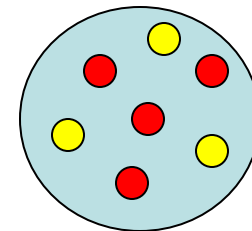
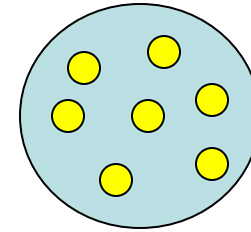
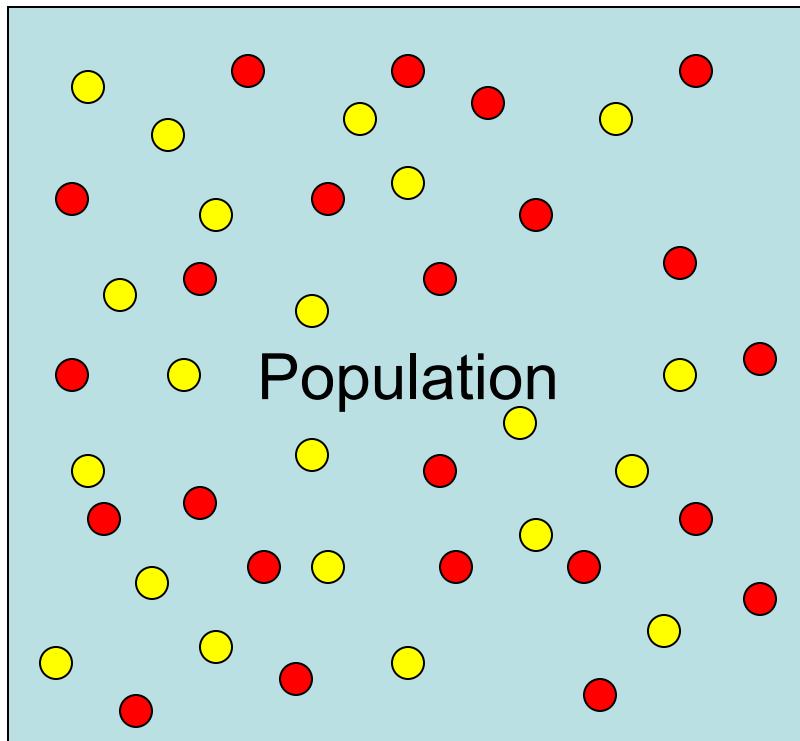
a) Population: The set of target objects of study

- *Socio-demographic study: all citizens of a given country*
- *Forestry survey: All trees in a study region*
- *Quality control: All product issues of a factory*

b) Sample

A reasonable small amount of individuals picked out from a given population for a specific study



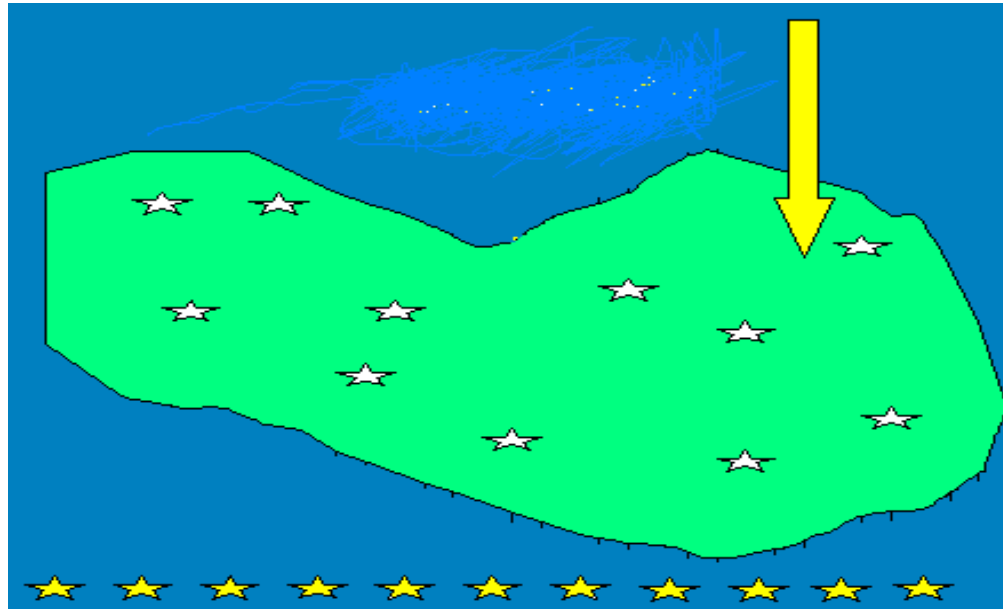


How to do the sampling

- **Representative for the population of study**
- **Corresponding to study target**

Sampling models

A. One sample model



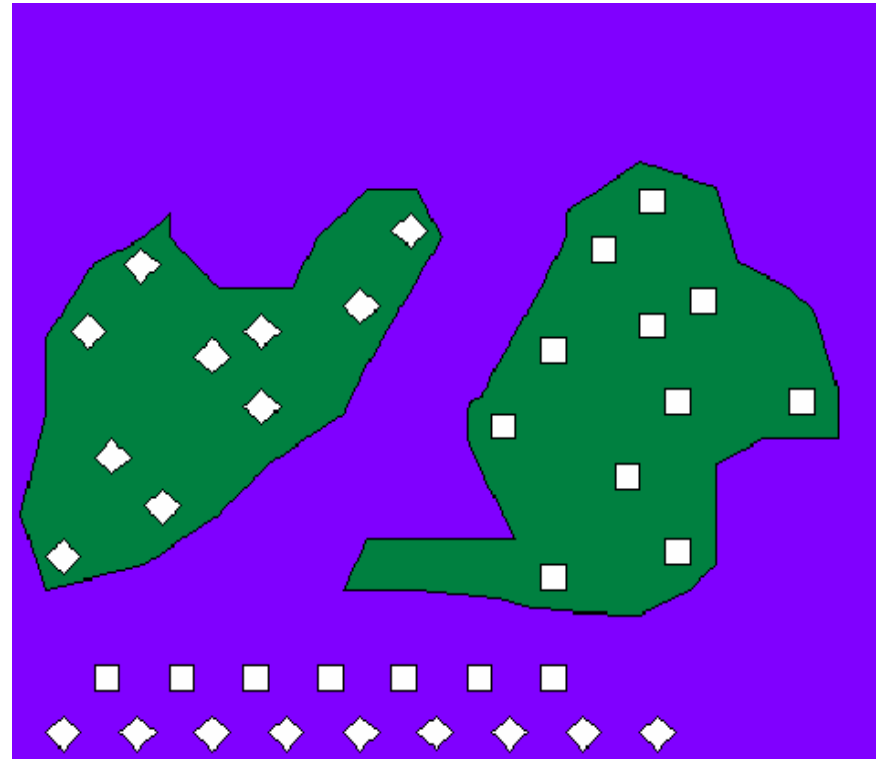
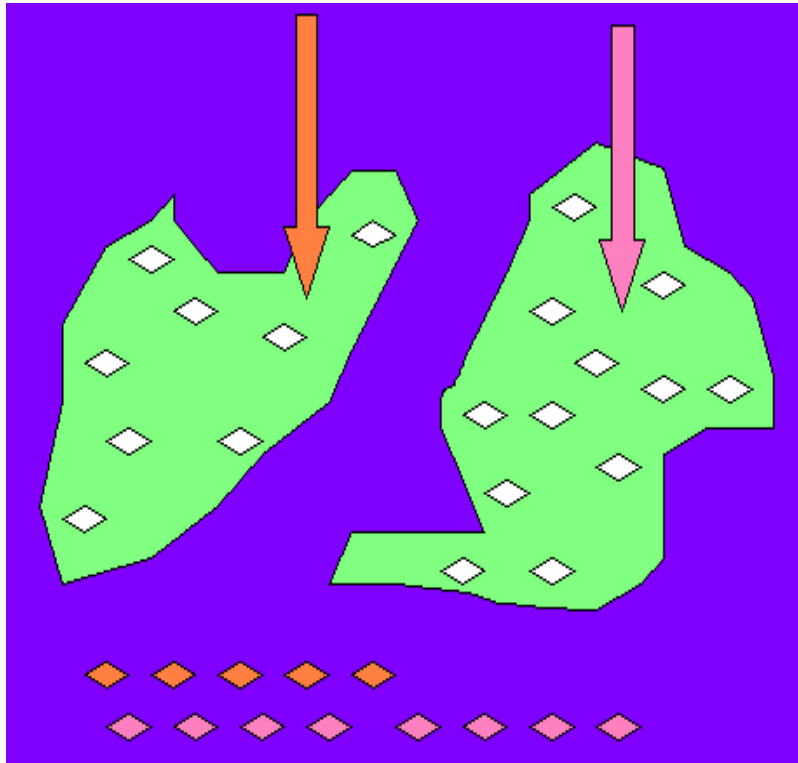
- One sample model usually concerns with an intervention on population: If the intervention should make some change in population?
- **Choose individuals from the population randomly to perform a sample**

■ **Example 1.** If in Ha Dong more than 90% motorcyclists use helmets?

■ **Example 2.** If proportion of girl-students less than 50% ?

■ **Example 3.** If in Viet Nam bread feeding is popular among more than 70% women?

B. Two independent samples model



Model of two groups of objects with different

- a) Intervention levels,
- b) Individual proper

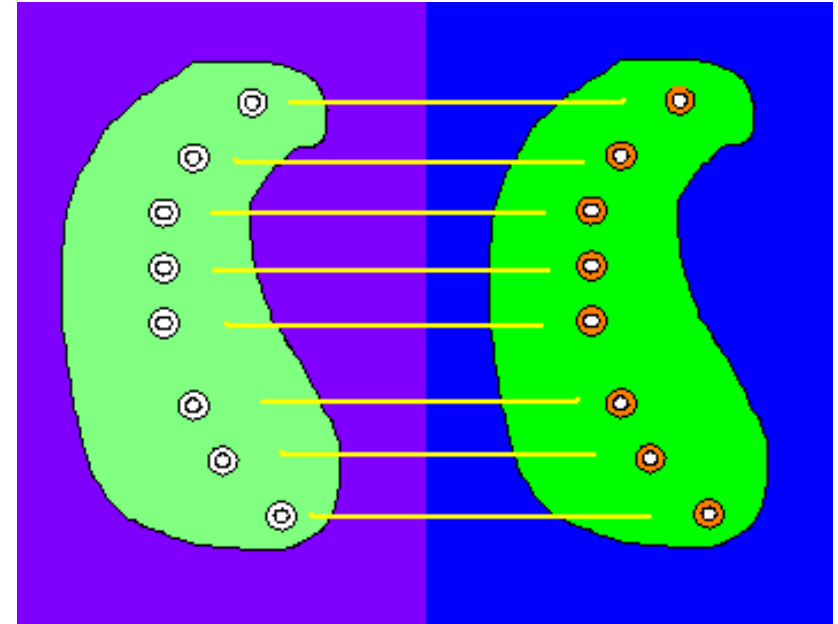
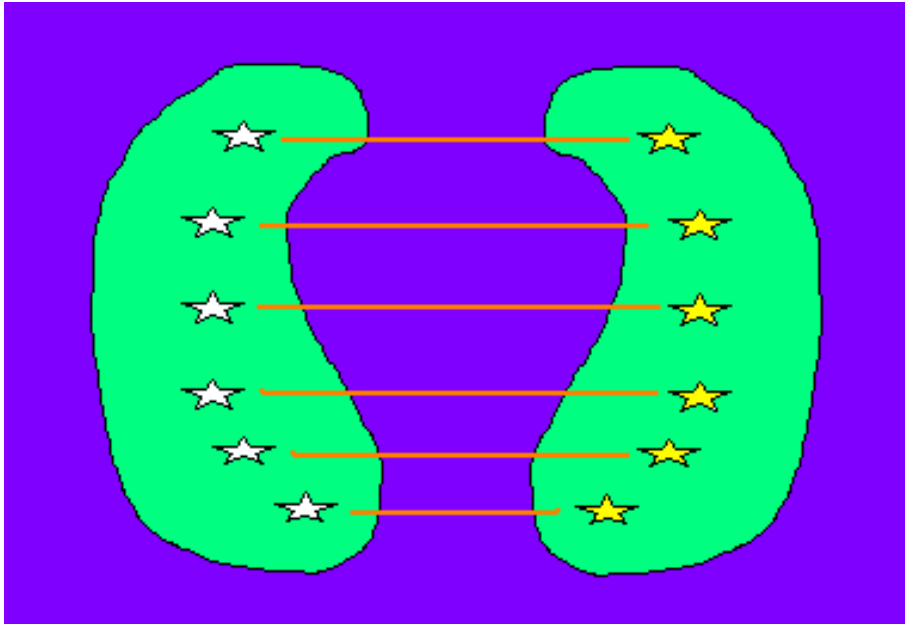
- **Example 1.** If women are better in foreign languages than men?
- **Example 2.** If there is any difference between Ha Noi and Ho Chi Minh City in immigration from rural areas?
- **Example 3.** If quality of coffee produced in Lam Dong is different than that in Dak Lak?

Notes:

In the model of two independent samples

- a) Observations numbers in two groups (sample sizes) may be different
- b) Observations of each group are independent from those of the second group**
- c) Sampling: Observations must be randomly selected from each of two groups**

C. Model of two dependent (paired) samples



- Two dependent samples model is used in a study when
- A) Each object in the first sample is chosen together with a **similar** (paired) object in the second sample, or
- B) Any object in the second sample is the **same** one in the first sample, but the measures in the two samples are taken under **different conditions**.

Notes: In the model of two paired samples

A) Observations amounts (sample sizes) of two samples are equal

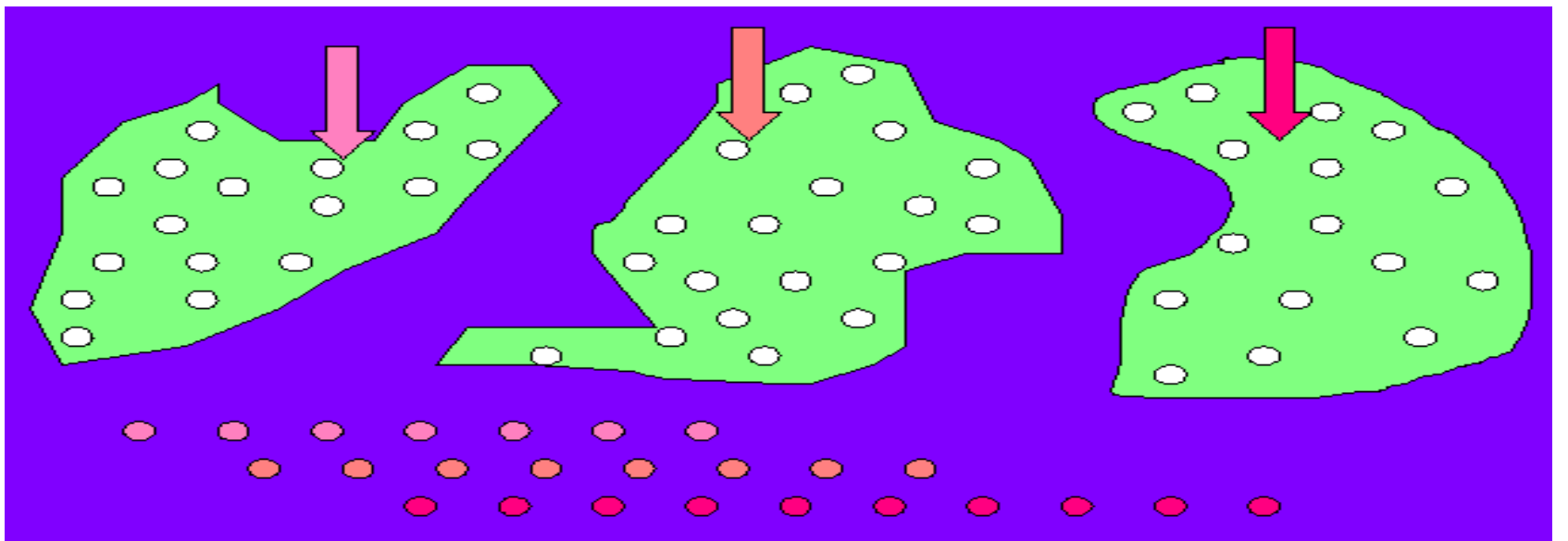
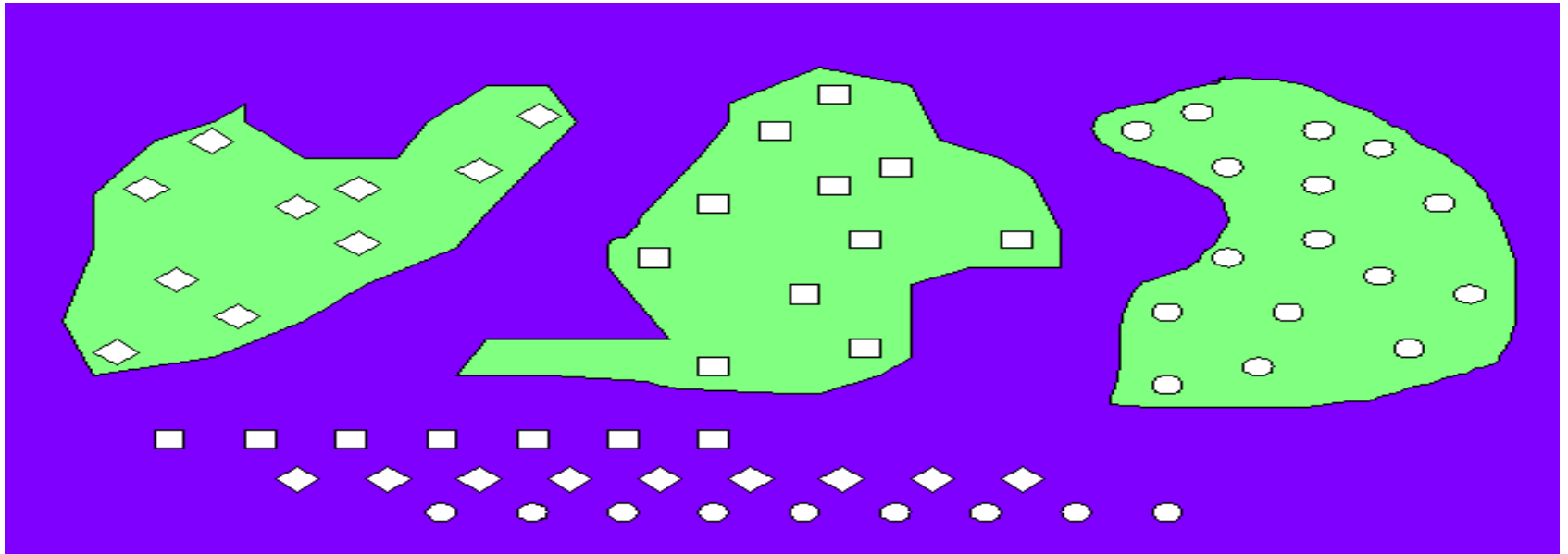
B) Information taken from one observation is related with that of correspondingly paired observation

C) In pairing to perform the samples, all factors which may influence on study issues must be taken into account

Example 1. To investigate the influence of cigarette on hypertension disease: perform two samples of smoking and nonsmoking people, each person from the nonsmoking group is paired with one smoking person similar about age, sex, weight, height, occupation, etc.

Example 2. Comparing 2022 and 2023 there is a changing in persons' opinion about Covid Vaccination?

D. Model of multi-independent samples



Notes:

In the model of multi- independent samples

- a) Observations numbers in groups (sample sizes) may be different
- b) Observations of each group are independent from those of other groups**
- c) Sampling: Observations must be randomly selected from each of groups**

- **Example 1.** Compare examination results of several high schools in Ha Noi
- **Example 2.** Compare salary in different economic sectors
- **Example 3.** Water supplying of ethnic groups?

Statistical data

DATA: Information, usually numerical or categorical

Elements, Variables, and Observations in statistical data:

- Elements (study units, objects) are the entities on which data are collected.
- A variable is a characteristic of interest for the elements.
- The set of variables' measurements obtained for a particular element is called an observation.
- A data set with n elements contains n observations.

Variable types

a) Quantitative variables (measures)

Example 1: Weight,
Temperature,
Density of a chemical substance in water

Example 2: Income
Salary
Price

Example 3: Age
Amount of children in family

Notice:

- Quantitative variable data indicate how many or how much.
- Quantitative variable data are always numeric.

b) Qualitative variables (nominal or categorical variables)

Characteristics of study object, usually with non-number values

Example: Sex (male-female), Residence place

Reason of borrow (for Health care, for Education, etc.

Occupation (Farmer, Worker, Vender

Transport (by foot, by boat, bicycle, motorbike, car, etc.)

– ***Ordinal qualitative variables:***

Values of variable can be ordered in certain way, presenting their importance levels.

Example: Housing, Water source, Transport mean, etc.

– ***Unordered qualitative variables:*** (nominal variables)

Values of variable can not be ranged in order

Example: Ethnic, Occupation, Reason of migration, etc.

CODING

Turning collected information into numerical form suitable for computing process

i) *Coding quantitative variables*

Values of quantitative variables are measures

→ The measures are taken directly as codes of variables

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ii) Coding qualitative variables

- *For ordered qualitative variables:*

Take integer numbers as codes for ordered levels of a given variable

- *For unordered qualitative variables:*

- + 1-st way : *Coding in the same way as for ordered variables,*
Each value of variable → one integer number

- + 2-nd way: From a given variable perform new auxiliary binary variables (impuls variables), each of those takes only two values 0 -1

Example:

a) Coding ordered qualitative variables

“Transport means”

- ~ By foot → 0
- ~ By bicycle → 1
- ~ By motorbike → 2

“Housing”

- ~ Homeless → 0
- ~ Thatched house → 1
- ~ Wooden house → 3
- ~ Apartment → 5
- ~ Villa → 6

b) Coding unordered qualitative variables

“Debt reason”: Production, Shoping, Health care, Education, Wedding

1-st way:

- ~ Production → 1
- ~ Shoping → 2
- ~ Health care → 3
- ~ Education → 4
- ~ Wedding → 5

2-nd way : Form up 5 auxiliary binary variables

Main variable values	Variable 1 Production	Variable 2 Shopping	Variable 3 Health care	Variable 4 Education	Variable 5 Wedding
Production	1	0	0	0	0
Shopping	0	1	0	0	0
Health care	0	0	1	0	0
Education	0	0	0	1	0
Wedding	0	0	0	0	1

4. Organizing data

Data matrix:

- Columns → **variables**,
- Rows → **Observations**

Example: Demographic survey

	Name	Age	Sex	Income	Height	Weight	Whatching TV	Housing
Person1	<i>Vân</i>	<i>27</i>	<i>Female</i>	<i>650000</i>	<i>1m55</i>	<i>55Kg</i>	<i>Every day</i>	<i>Hired</i>
Person 2	<i>Bường</i>	<i>46</i>	<i>Male</i>	<i>980000</i>	<i>1m68</i>	<i>67Kg</i>	<i>Rarely</i>	<i>Brick H.</i>
. . .	<i>...</i>	<i>. . .</i>	<i>. . .</i>	<i>. . .</i>	<i>. . .</i>	<i>. . .</i>	<i>. . .</i>	<i>. . .</i>
Person 40	<i>Việt</i>	<i>31</i>	<i>Male</i>	<i>775000</i>	<i>1m73</i>	<i>58Kg</i>	<i>Every day</i>	<i>Wooden</i>
Person 41	<i>Canh</i>	<i>77</i>	<i>Female</i>	<i>325000</i>	<i>1m49</i>	<i>46Kg</i>	<i>Never</i>	<i>Thatched</i>



1	VAN	27	2	650	1.55	55	2	0
2	BUONG	46	1	980	1.68	67	1	5
...	...	...	...	...	...	...	...	...
40	VIET	31	1	775	1.73	58	2	3
41	CANH	77	2	325	1.49	46	0	1